

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

**Product Identifier**

**Product Description:** Borane dimethyl sulfide complex  
**Cat No. :** 177060000; 177061000; 177068000  
**Synonyms** BMS; Dimethyl sulfideborane  
**Molecular Formula** C2 H9 B S

**Relevant identified uses of the substance or mixture and uses advised against**

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

**Details of the supplier of the safety data sheet**

|                                                                                                                                                                                               |                                                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <b>Importer</b><br>Fisher Scientific Korea<br>D5,D6, Incheon Airport Logistics Complex<br>150, Gonghangdong-Ro 296 Beon-Gil<br>Jung-Gu, Incheon<br>Tel: +82-1661-9555<br>Fax: +82-2-2023-0603 | <b>Supplier</b><br>Acros Organics BVBA<br>Janssen Pharmaceutica laan 3a<br>2440 Geel, Belgium |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|

**E-mail address** Chem.KR@thermofisher.com

**Emergency Telephone Number**

Emergency telephone: Medical: +(82) 070-7686-0086 or + 1-703-741-5970  
 CHEMTREC: 080 822 1374 (Local), CHEMTREC: 1-800-424-9300 or + 1-703-527-3887  
 Korea: 00-308-13-2549 (24 hours a day, 7 days a week)

## SECTION 2: HAZARDS IDENTIFICATION

**Classification of the substance or mixture**
**Physical hazards**

|                                                                        |            |
|------------------------------------------------------------------------|------------|
| Flammable liquids                                                      | Category 2 |
| Substances/mixtures which, in contact with water, emit flammable gases | Category 1 |

**Health hazards**

|                                   |            |
|-----------------------------------|------------|
| Acute Oral Toxicity               | Category 3 |
| Acute Dermal Toxicity             | Category 3 |
| Skin Corrosion/Irritation         | Category 2 |
| Serious Eye Damage/Eye Irritation | Category 1 |
| Reproductive Toxicity             | Category 1 |

**Environmental hazards**

|                          |            |
|--------------------------|------------|
| Chronic aquatic toxicity | Category 3 |
|--------------------------|------------|

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## Label Elements



**Signal Word**

**Danger**

### **Hazard Statements**

H225 - Highly flammable liquid and vapor  
H260 - In contact with water releases flammable gases which may ignite spontaneously  
H301 + H311 - Toxic if swallowed or in contact with skin  
H315 - Causes skin irritation  
H318 - Causes serious eye damage  
H360 - May damage fertility or the unborn child  
H412 - Harmful to aquatic life with long lasting effects

### **Precautionary Statements**

#### **Prevention**

P201 - Obtain special instructions before use  
P202 - Do not handle until all safety precautions have been read and understood  
P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P223 - Do not allow contact with water  
P231 + P232 - Handle and store contents under inert gas. Protect from moisture  
P233 - Keep container tightly closed  
P240 - Ground and bond container and receiving equipment  
P241 - Use explosion-proof electrical/ ventilating/ lighting equipment  
P242 - Use non-sparking tools  
P243 - Take action to prevent static discharges  
P264 - Wash hands and face thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P273 - Avoid release to the environment

#### **Response**

P301 + P310 - IF SWALLOWED: Immediately call a POISON CENTER or doctor  
P302 + P335 + P334 - IF ON SKIN: Brush off loose particles from skin. Immerse in cool water or wrap in wet bandages  
P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P308 + P313 - IF exposed or concerned: Get medical advice/attention  
P310 - Immediately call a POISON CENTER or doctor  
P312 - Call a POISON CENTER or doctor if you feel unwell  
P321 - Specific treatment (see supplemental first aid instructions on this label)  
P330 - Rinse mouth  
P332 + P313 - If skin irritation occurs: Get medical advice/attention  
P361 + P364 - Take off immediately all contaminated clothing and wash it before reuse  
P362 + P364 - Take off contaminated clothing and wash it before reuse  
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam for extinction

#### **Storage**

P403 + P235 - Store in a well-ventilated place. Keep cool  
P402 + P404 - Store in a dry place. Store in a closed container  
P405 - Store locked up

#### **Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

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## Other Hazards

Reacts violently with water

This product does not contain any known or suspected endocrine disruptors

Toxic to terrestrial vertebrates

## NFPA

Health  
3

Flammability  
3

Instability  
2

Physical hazards  
W

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### 3.1. Substances

| Component                                  | Common Name                 | CAS No     | Index No   | Weight %  |
|--------------------------------------------|-----------------------------|------------|------------|-----------|
| Boron, trihydro[thiobis[methane]]-, (T-4)- | BMS; Dimethyl sulfideborane | 13292-87-0 | 2008-1-560 | 90 - 95   |
| Dimethyl sulfide                           | Dimethyl sulfide            | 75-18-3    | KE-33766   | >=5 - <10 |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### **General Advice**

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

#### **Eye Contact**

In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

#### **Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

#### **Ingestion**

Do NOT induce vomiting. Call a physician or poison control center immediately.

#### **Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.

#### **Self-Protection of the First Aider**

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

### Most important symptoms and effects, both acute and delayed

Causes severe eye damage. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting. Gastrointestinal discomfort.

### Indication of any immediate medical attention and special treatment needed

#### **Notes to Physician**

Treat symptomatically. Symptoms may be delayed.

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### **Suitable Extinguishing Media**

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam. Water mist may be used to cool closed containers.

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## Extinguishing media which must not be used for safety reasons

Water.

## Special hazards arising from the substance or mixture

Flammable. Reacts violently with water. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

## Hazardous Combustion Products

Hydrogen, Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Sulfur oxides, Oxides of boron, Hydrogen chloride gas.

## Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Ensure adequate ventilation. Use personal protective equipment as required. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas. Remove all sources of ignition. Take precautionary measures against static discharges.

### Environmental precautions

Should not be released into the environment. Do not flush into surface water or sanitary sewer system.

### Methods and Material for Containment and Cleaning Up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Do not expose spill to water. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Do not allow contact with water. Handle under an inert atmosphere. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges.

### Conditions for Safe Storage, Including any Incompatibilities

Refrigerator/flammables. Keep under nitrogen. Keep away from heat, sparks and flame. Keep from any possible contact with water. Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from water or moist air. Protect from moisture.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component | CAS No     | Korea      | ACGIH TLV  | OSHA PEL   |
|-----------|------------|------------|------------|------------|
| Boron,    | 13292-87-0 | Not listed | Not listed | Not listed |

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|                                        |         |            |             |            |
|----------------------------------------|---------|------------|-------------|------------|
| trihydro[thiobis[methane]]-,<br>(T-4)- |         |            |             |            |
| Dimethyl sulfide                       | 75-18-3 | Not listed | TWA: 10 ppm | Not listed |

| Component                                        | CAS No     | European Union | The United Kingdom | Germany    |
|--------------------------------------------------|------------|----------------|--------------------|------------|
| Boron,<br>trihydro[thiobis[methane]]-,<br>(T-4)- | 13292-87-0 | Not listed     | Not listed         | Not listed |
| Dimethyl sulfide                                 | 75-18-3    | Not listed     | Not listed         | Not listed |

## ACGIH - Biological Exposure Indices

| Component                                        | CAS No     | ACGIH - Biological Exposure Indices |
|--------------------------------------------------|------------|-------------------------------------|
| Boron,<br>trihydro[thiobis[methane]]-,<br>(T-4)- | 13292-87-0 | Not listed                          |
| Dimethyl sulfide                                 | 75-18-3    | Not listed                          |

## Exposure Controls

### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting equipment.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles

#### Hand Protection

Protective gloves

#### Skin and body protection

Long sleeved clothing

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

### Personal protective equipment

#### Respiratory Protection

Use only those certified by the Korea Occupational Safety and Health Administration.

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

#### Recommended Filter type:

low boiling organic solvent Type AX Brown conforming to EN371 or Organic gases and vapours filter Type A Brown conforming to EN14387

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

When RPE is used a face piece Fit Test should be conducted

## Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

## Environmental exposure controls

Prevent product from entering drains

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

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|                                          |                               |                                   |
|------------------------------------------|-------------------------------|-----------------------------------|
| Appearance (Physical State, Color, etc.) | Amber Liquid                  |                                   |
| Odor                                     | pungent                       |                                   |
| Odor Threshold                           | No data available             |                                   |
| pH                                       | No information available      |                                   |
| Melting Point/Range                      | -40 - -37 °C / -40 - -34.6 °F |                                   |
| Softening Point                          | No data available             |                                   |
| Boiling Point/Range                      | No information available      |                                   |
| Flash Point                              | 3 °C / 37.4 °F                | Method - No information available |
| Evaporation Rate                         | No data available             |                                   |
| Flammability (solid,gas)                 | Not applicable                | Liquid                            |
| Explosion Limits                         | No data available             |                                   |
| Vapor Pressure                           | 19.1 mmHg @ 22.2 °C           |                                   |
| Vapor Density                            | No information available      | (Air = 1.0)                       |
| Specific Gravity / Density               | 0.790                         |                                   |
| Bulk Density                             | Not applicable                | Liquid                            |
| Water Solubility                         | Reacts violently with water   |                                   |
| Solubility in other solvents             | No information available      |                                   |

## Partition Coefficient (n-octanol/water)

| Component                                  | CAS No     | log Pow           |
|--------------------------------------------|------------|-------------------|
| Boron, trihydro[thiobis[methane]]-, (T-4)- | 13292-87-0 | No data available |
| Dimethyl sulfide                           | 75-18-3    | 0.84              |

|                           |                          |                                             |
|---------------------------|--------------------------|---------------------------------------------|
| Autoignition Temperature  | 91 °C / 195.8 °F         |                                             |
| Decomposition Temperature | 44 °C                    |                                             |
| Viscosity                 | No data available        |                                             |
| Explosive Properties      |                          | Vapors may form explosive mixtures with air |
| Oxidizing Properties      | No information available |                                             |
| Molecular Formula         | C2 H9 B S                |                                             |
| Molecular Weight          | 75.95                    |                                             |

## SECTION 10: STABILITY AND REACTIVITY

|                                           |                                                                                          |
|-------------------------------------------|------------------------------------------------------------------------------------------|
| <u>Reactivity</u>                         | Yes                                                                                      |
| <u>Chemical Stability</u>                 | Reacts violently with water, liberating extremely flammable gases. Moisture sensitive.   |
| <u>Possibility of Hazardous Reactions</u> |                                                                                          |
| Hazardous Polymerization                  | Hazardous polymerization does not occur.                                                 |
| Hazardous Reactions                       | None under normal processing. Reacts violently with water.                               |
| <u>Conditions to Avoid</u>                | Keep away from open flames, hot surfaces and sources of ignition. Incompatible products. |

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Exposure to moist air or water. Exposure to moisture.

## Incompatible Materials

Acids. Water. Alcohols. Acid anhydrides. Acid chlorides.

## Hazardous Decomposition Products

Hydrogen. Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Sulfur oxides. Oxides of boron. Hydrogen chloride gas.

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

#### Information on expected route of exposure

Use only in closed systems permitted

**Inhalation** Not an expected route of exposure.  
**Ingestion** May be harmful if swallowed.  
**Eyes** Avoid contact with eyes. Corrosive to the eyes and may cause severe damage including blindness. May cause irritation. Risk of serious damage to eyes.  
**Skin** Avoid contact with skin. Skin Corrosion/Irritation. May cause irritation. Contact with moist skin may cause skin burns. Harmful in contact with skin.

#### Information on Health Hazards

##### (a) acute toxicity;

**Oral** Category 3  
**Dermal** Category 3  
**Inhalation** Based on available data, the classification criteria are not met

#### Toxicology data for the components

| Component                                  | CAS No     | LD50 Oral            | LD50 Dermal            | LC50 Inhalation              |
|--------------------------------------------|------------|----------------------|------------------------|------------------------------|
| Boron, trihydro[thiobis[methane]]-, (T-4)- | 13292-87-0 | <500 mg/kg (Rat)     | >2000 mg/kg (Rabbit)   | No data available            |
| Dimethyl sulfide                           | 75-18-3    | > 2000 mg/kg ( Rat ) | >5000 mg/kg ( Rabbit ) | LC50 = 40250 ppm ( Rat ) 4 h |

(b) skin corrosion/irritation; Category 2

(c) serious eye damage/irritation; Category 1

##### (d) respiratory or skin sensitization;

**Respiratory** No data available  
**Skin** No data available

| Component                                  | CAS No     | Test method       | Test species      | Study result      |
|--------------------------------------------|------------|-------------------|-------------------|-------------------|
| Boron, trihydro[thiobis[methane]]-, (T-4)- | 13292-87-0 | No data available | No data available | No data available |
| Dimethyl sulfide                           | 75-18-3    | No data available | No data available | No data available |

(e) germ cell mutagenicity; No data available

| Component                                  | CAS No     | Test method       | Test species      | Study result      |
|--------------------------------------------|------------|-------------------|-------------------|-------------------|
| Boron, trihydro[thiobis[methane]]-, (T-4)- | 13292-87-0 | No data available | No data available | No data available |
| Dimethyl sulfide                           | 75-18-3    | No data available | No data available | No data available |

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(f) carcinogenicity; No data available

| Component                                  | CAS No     | Test method       | Test species / Duration | Study result      |
|--------------------------------------------|------------|-------------------|-------------------------|-------------------|
| Boron, trihydro[thiobis[methane]]-, (T-4)- | 13292-87-0 | No data available | No data available       | No data available |
| Dimethyl sulfide                           | 75-18-3    | No data available | No data available       | No data available |

There are no known carcinogenic chemicals in this product

| Component                                  | CAS No     | IARC       | NTP        | ACGIH      | OSHA       | UK         |
|--------------------------------------------|------------|------------|------------|------------|------------|------------|
| Boron, trihydro[thiobis[methane]]-, (T-4)- | 13292-87-0 | Not listed | Not listed | Not listed | Not listed | Not listed |
| Dimethyl sulfide                           | 75-18-3    | Not listed | Not listed | Not listed | Not listed | Not listed |

(g) reproductive toxicity; Category 1

| Component                                  | CAS No     | Test method       | Test species / Duration | Study result      |
|--------------------------------------------|------------|-------------------|-------------------------|-------------------|
| Boron, trihydro[thiobis[methane]]-, (T-4)- | 13292-87-0 | No data available | No data available       | No data available |
| Dimethyl sulfide                           | 75-18-3    | No data available | No data available       | No data available |

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs None known.

(j) aspiration hazard; No data available

## Other Adverse Effects

Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting. Gastrointestinal discomfort.

| Component                                  | CAS No     | EU - Endocrine Disruptors Candidate List | EU - Endocrine Disruptors - Evaluated Substances | Japan - Endocrine Disruptor Information |
|--------------------------------------------|------------|------------------------------------------|--------------------------------------------------|-----------------------------------------|
| Boron, trihydro[thiobis[methane]]-, (T-4)- | 13292-87-0 | Not applicable                           | Not applicable                                   | Not applicable                          |
| Dimethyl sulfide                           | 75-18-3    | Not applicable                           | Not applicable                                   | Not applicable                          |

## SECTION 12: ECOLOGICAL INFORMATION

### Ecotoxicity effects

Do not empty into drains. Harmful to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment. Reacts with water so no ecotoxicity data for the substance is available.

| Component                                  | CAS No     | Freshwater Fish                                         | Water Flea                           | Freshwater Algae  | Microtox          |
|--------------------------------------------|------------|---------------------------------------------------------|--------------------------------------|-------------------|-------------------|
| Boron, trihydro[thiobis[methane]]-, (T-4)- | 13292-87-0 | No data available                                       | No data available                    | No data available | No data available |
| Dimethyl sulfide                           | 75-18-3    | LC50: = 213 mg/L, 96h semi-static (Oncorhynchus mykiss) | EC50: = 23 mg/L, 48h (Daphnia pulex) | No data available | No data available |

Persistence and degradability No information available



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|                                              |                                                                                                                                              |
|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Persistence</b>                           | Persistence is unlikely, based on information available.                                                                                     |
| <b>Degradability</b>                         | Reacts with water.                                                                                                                           |
| <b>Degradation in sewage treatment plant</b> | Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants. Reacts violently with water. |

**Bioaccumulative potential** Product does not bioaccumulate due to reaction with water

| Component        | log Pow | Bioconcentration factor (BCF) |
|------------------|---------|-------------------------------|
| Dimethyl sulfide | 0.84    | No data available             |

**Mobility in soil** Reacts violently with water . Is not likely mobile in the environment.

## Ozone Depletion Potential

| Component                                  | CAS No     | Ozone Depletion Potential |
|--------------------------------------------|------------|---------------------------|
| Boron, trihydro[thiobis[methane]]-, (T-4)- | 13292-87-0 | Not listed                |
| Dimethyl sulfide                           | 75-18-3    | Not listed                |

**Other adverse effects** No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

### Waste treatment methods

**Waste from Residues/Unused Products** Waste is classified as hazardous. Dispose in accordance with the Wastes Control Act (폐기물관리법).

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

**Other Information** Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations. Do not empty into drains. Do not let this chemical enter the environment.

## SECTION 14: TRANSPORT INFORMATION

### Road and Rail Transport

|                                |                                                              |
|--------------------------------|--------------------------------------------------------------|
| <b>UN-No</b>                   | UN3399                                                       |
| <b>Proper Shipping Name</b>    | ORGANOMETALLIC SUBSTANCE, LIQUID, WATER-REACTIVE, FLAMMABLE  |
| <b>Technical Shipping Name</b> | Boron, trihydro[thiobis[methane]]-, (T-4)-, Dimethyl sulfide |
| <b>Hazard Class</b>            | 4.3                                                          |
| <b>Subsidiary Hazard Class</b> | 3                                                            |
| <b>Packing Group</b>           | I                                                            |

### IATA

|                                |                                                              |
|--------------------------------|--------------------------------------------------------------|
| <b>UN-No</b>                   | UN3399                                                       |
| <b>Proper Shipping Name</b>    | ORGANOMETALLIC SUBSTANCE, LIQUID, WATER-REACTIVE, FLAMMABLE  |
| <b>Technical Shipping Name</b> | Boron, trihydro[thiobis[methane]]-, (T-4)-, Dimethyl sulfide |
| <b>Hazard Class</b>            | 4.3                                                          |
| <b>Subsidiary Hazard Class</b> | 3                                                            |
| <b>Packing Group</b>           | I                                                            |

### IMDG/IMO

|                                |                                                              |
|--------------------------------|--------------------------------------------------------------|
| <b>UN-No</b>                   | UN3399                                                       |
| <b>Proper Shipping Name</b>    | ORGANOMETALLIC SUBSTANCE, LIQUID, WATER-REACTIVE, FLAMMABLE  |
| <b>Technical Shipping Name</b> | Boron, trihydro[thiobis[methane]]-, (T-4)-, Dimethyl sulfide |
| <b>Hazard Class</b>            | 4.3                                                          |
| <b>Subsidiary Hazard Class</b> | 3                                                            |
| <b>Packing Group</b>           | I                                                            |
| <b>Marine Pollutant</b>        | No hazards identified                                        |

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Special Precautions for User No special precautions required

## SECTION 15: REGULATORY INFORMATION

Safety, health and environmental regulations/legislation specific for the substance or mixture

Legend: X - Listed '-' - Not Listed

### International Inventories

| Component                                  | CAS No     | KECL       | TSCA | EINECS    | IECSC | DSL | NDSL | PICCS | ENCs | ISHL | AICS |
|--------------------------------------------|------------|------------|------|-----------|-------|-----|------|-------|------|------|------|
| Boron, trihydro[thiobis(methane)]-, (T-4)- | 13292-87-0 | 2008-1-560 | X    | 236-313-6 | X     | -   | X    | -     | -    | X    | -    |
| Dimethyl sulfide                           | 75-18-3    | KE-33766   | X    | 200-846-2 | X     | X   | -    | X     | X    | X    | X    |

| Component                                  | CAS No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|--------------------------------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Boron, trihydro[thiobis(methane)]-, (T-4)- | 13292-87-0 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Dimethyl sulfide                           | 75-18-3    | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

| Component                                  | CAS No     | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential |
|--------------------------------------------|------------|----------------|------------------------------|---------------------------|
| Boron, trihydro[thiobis(methane)]-, (T-4)- | 13292-87-0 | Not applicable | Not applicable               | Not applicable            |
| Dimethyl sulfide                           | 75-18-3    | Listed         | Not applicable               | Not applicable            |

### Korean National Regulations

| Component                                  | CAS No     | Act on Registration and Evaluation of Chemical Substances (K-REACH) | Authorised Chemicals | Existing Substances Subject to Registration |
|--------------------------------------------|------------|---------------------------------------------------------------------|----------------------|---------------------------------------------|
| Boron, trihydro[thiobis(methane)]-, (T-4)- | 13292-87-0 | Annex 2 - 2008-1-560                                                | Not applicable       | Not applicable                              |
| Dimethyl sulfide                           | 75-18-3    | Annex 1 - KE-33766                                                  | Not applicable       | Not applicable                              |

| Component                                  | CAS No     | Chemical Control Act - Toxic Chemicals | Chemical Control Act - Prohibited Chemicals | Chemical Control Act - Use Restricted Chemicals |
|--------------------------------------------|------------|----------------------------------------|---------------------------------------------|-------------------------------------------------|
| Boron, trihydro[thiobis(methane)]-, (T-4)- | 13292-87-0 | 2008-1-0560 (>25%)                     | Not applicable                              | Not applicable                                  |
| Dimethyl sulfide                           | 75-18-3    | Not applicable                         | Not applicable                              | Not applicable                                  |

| Component                                  | CAS No     | Chemical Control Act - Accident Precaution Chemicals (% in mixtures) | Chemical Control Act - Accident Precaution Chemicals - Quantity limits Storage (% in mixtures) | Chemical Control Act - Accident Precaution Chemicals - Quantity limits Manufacture/Use (% in mixtures) |
|--------------------------------------------|------------|----------------------------------------------------------------------|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Boron, trihydro[thiobis(methane)]-, (T-4)- | 13292-87-0 | Not applicable                                                       | Not applicable                                                                                 | Not applicable                                                                                         |
| Dimethyl sulfide                           | 75-18-3    | Not applicable                                                       | Not applicable                                                                                 | Not applicable                                                                                         |

| Component                                  | CAS No     | Waste Control Law | Ministry of Environment - CMR risk | Ministry of Environment - Critically Controlled Substance |
|--------------------------------------------|------------|-------------------|------------------------------------|-----------------------------------------------------------|
| Boron, trihydro[thiobis(methane)]-, (T-4)- | 13292-87-0 | > 25% (CCA)       | Not applicable                     | Not applicable                                            |

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|                            |         |                |                |                |
|----------------------------|---------|----------------|----------------|----------------|
| (T-4)-<br>Dimethyl sulfide | 75-18-3 | Not applicable | Not applicable | Not applicable |
|----------------------------|---------|----------------|----------------|----------------|

CCA = Chemical Control Act

| Component                                  | CAS No     | ISHA - Harmful Agents Subject to Work Environment Monitoring | ISHA - Prohibited substances | ISHA - Substances requiring permission |
|--------------------------------------------|------------|--------------------------------------------------------------|------------------------------|----------------------------------------|
| Boron, trihydro[thiobis[methane]]-, (T-4)- | 13292-87-0 | Not applicable                                               | Not applicable               | Not applicable                         |
| Dimethyl sulfide                           | 75-18-3    | Not applicable                                               | Not applicable               | Not applicable                         |

| Component                                  | CAS No     | ISHA - Substances subject to control | ISHA - Harmful Agents Requiring Health Examination | ISHA - Permissible Exposure Limits |
|--------------------------------------------|------------|--------------------------------------|----------------------------------------------------|------------------------------------|
| Boron, trihydro[thiobis[methane]]-, (T-4)- | 13292-87-0 | Not applicable                       | Not applicable                                     | Not applicable                     |
| Dimethyl sulfide                           | 75-18-3    | Not applicable                       | Not applicable                                     | Not applicable                     |

| Component                                  | CAS No     | ISHA - Subject to Process Safety Reports (minimum quantity) | ISHA - Threshold Limit Values (TLVs) Chemicals | ISHA - Special management materials |
|--------------------------------------------|------------|-------------------------------------------------------------|------------------------------------------------|-------------------------------------|
| Boron, trihydro[thiobis[methane]]-, (T-4)- | 13292-87-0 | 5000 kg                                                     | Not applicable                                 | Not applicable                      |
| Dimethyl sulfide                           | 75-18-3    | 5000 kg                                                     | Not applicable                                 | Not applicable                      |

## National Fire Association - Dangerous Substances Minimum quantity requiring a permit

| Component                                  | CAS No     | Class 1 - Oxidising solids | Class 2 - Flammable solid | Class 3 - Spontaneously Combustible Substances and Dangerous Substances When Wet | Class 4 - Flammable liquids            | Class 5 - Self-reactive substances | Class 6 - Oxidising liquids |
|--------------------------------------------|------------|----------------------------|---------------------------|----------------------------------------------------------------------------------|----------------------------------------|------------------------------------|-----------------------------|
| Boron, trihydro[thiobis[methane]]-, (T-4)- | 13292-87-0 | Not applicable             | Not applicable            | Not applicable                                                                   | 2. Group 1 Petroleum (Insoluble) 200 L | Not applicable                     | Not applicable              |
| Dimethyl sulfide                           | 75-18-3    | Not applicable             | Not applicable            | Not applicable                                                                   | 1. Special flammables 50 L             | Not applicable                     | Not applicable              |

## Control Parameters

| Component                                  | CAS No     | Korea      | ACGIH - Biological Exposure Indices |
|--------------------------------------------|------------|------------|-------------------------------------|
| Boron, trihydro[thiobis[methane]]-, (T-4)- | 13292-87-0 | Not listed | Not listed                          |
| Dimethyl sulfide                           | 75-18-3    | Not listed | Not listed                          |

## US Management Information

OSHA - Occupational Safety and Health Administration

Not applicable

| Component                                  | CAS No     | Specifically Regulated Chemicals | Highly Hazardous Chemicals |
|--------------------------------------------|------------|----------------------------------|----------------------------|
| Boron, trihydro[thiobis[methane]]-, (T-4)- | 13292-87-0 | Not applicable                   | Not applicable             |
| Dimethyl sulfide                           | 75-18-3    | Not applicable                   | Not applicable             |

CERCLA Not applicable

| Component                                  | CAS No     | CERCLA Extremely Hazardous Substances RQs | Hazardous Substances RQs | SARA 313 - Threshold Values % |
|--------------------------------------------|------------|-------------------------------------------|--------------------------|-------------------------------|
| Boron, trihydro[thiobis[methane]]-, (T-4)- | 13292-87-0 | Not applicable                            | Not applicable           | Not applicable                |

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|                  |         |                |                |                |
|------------------|---------|----------------|----------------|----------------|
| (T-4)-           |         |                |                |                |
| Dimethyl sulfide | 75-18-3 | Not applicable | Not applicable | Not applicable |

## GHS Classification - According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567

Danger.

H225 - Highly flammable liquid and vapor. H260 - In contact with water releases flammable gases which may ignite spontaneously. H301 + H311 - Toxic if swallowed or in contact with skin. H315 - Causes skin irritation. H318 - Causes serious eye damage. H360FD - May damage fertility. May damage the unborn child. H412 - Harmful to aquatic life with long lasting effects. EUH014 - Reacts violently with water.

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking. P231 + P232 - Handle and store contents under inert gas. Protect from moisture. P280 - Wear protective gloves/protective clothing/eye protection/face protection. P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower. P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing. P310 - Immediately call a POISON CENTER or doctor/physician.

## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.

Chemical incident response training.

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**Revision Date** 05-Mar-2025

**Revision Number** 7

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Revision Summary

SDS sections updated.

## MOEL's Public Notice No. 2023-9 (Standards for Classification and Labeling of Chemical Substances and Safety Data Sheets)

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**